

Accuracy Test Report

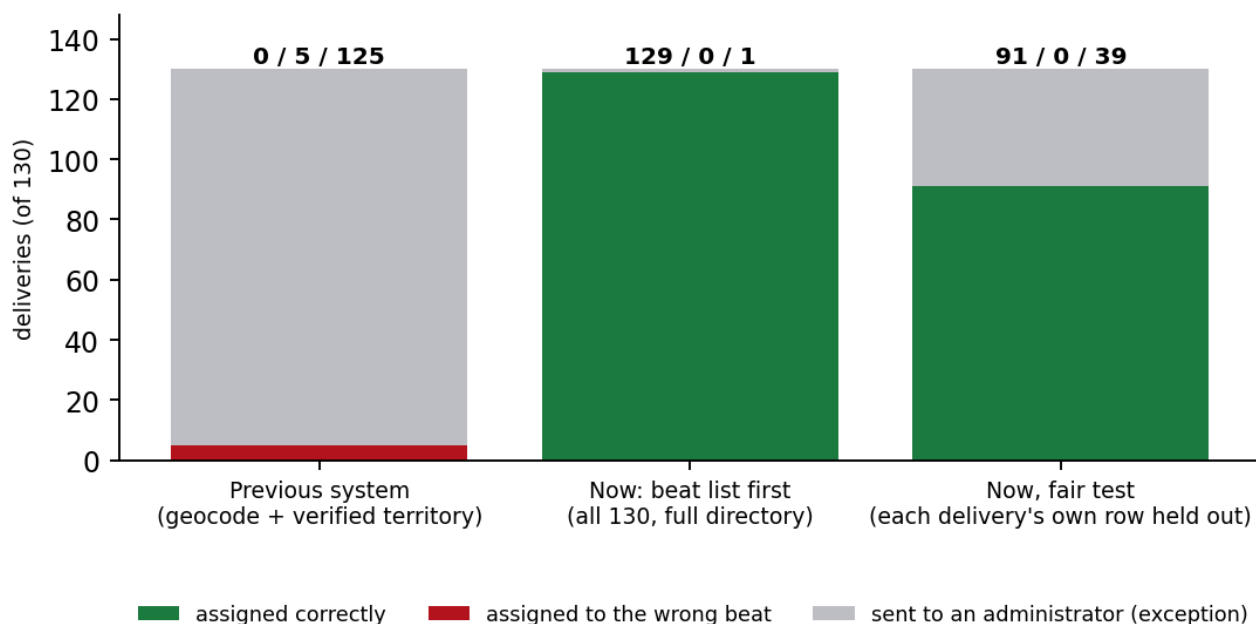
Address to beat matching, and the five route algorithms - Bhandup West data set, re-run after the address-matching, territory, geocoding, routing and app changes.

Read this first. The test file gives the answer key in its own layout (5 deliveries per beat, in beat order), and the beat directory supplied with it lists every delivery's own main area. So the full-directory figure is partly a lookup of the answer; the **fair figure is the hold-out** one (each delivery's own directory row removed). The coordinates the geocoder returned are OpenStreetMap anchors near the named places, not house positions: nothing here measures house-level location accuracy, and the geocode / territory methods never fired on this data (all 130 points are area- or pincode-level).

1. Address to beat assignment

The same 130 deliveries (deliveries.csv, unchanged), imported through the real import into a fresh database with the beat list imported through the beat-list wizard (26 beats, 201 locality records), Nominatim as the geocoder. Every number below is counted from what the system stored.

	Previous system	Now (all 130)	Now, fair test (hold-out)
Assigned correctly	0	129	91
Assigned to the wrong beat	5	0	0
Sent to an administrator (exception)	125	1	39
Accuracy of all 130	0.0 %	99.2 %	70.0 %
Accuracy of the ones assigned automatically	0 % (5 assigned, all wrong)	100 %	100 %



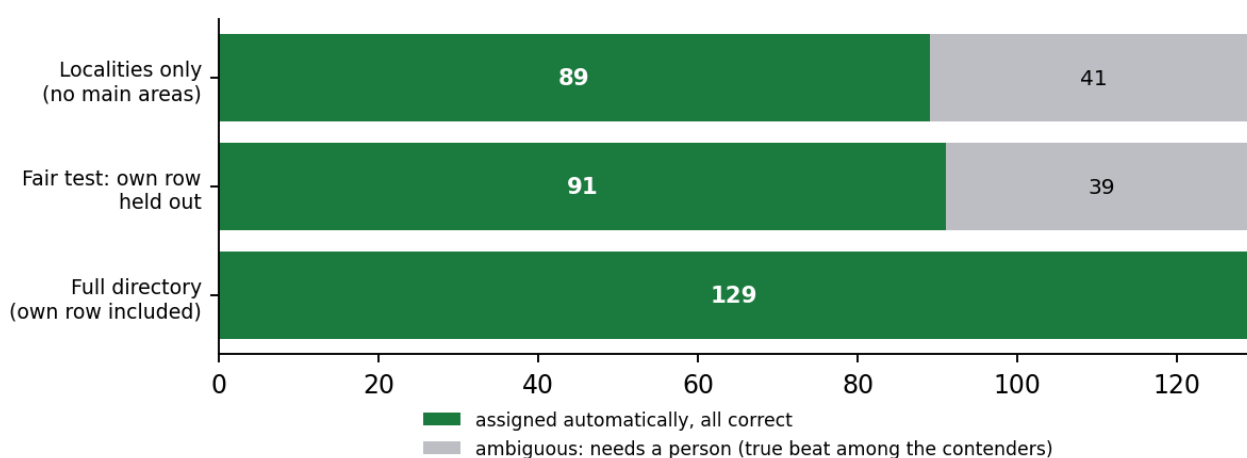
How the system decided

Method	Deliveries	Correct
NAME - the beat list identified the beat	129	129
NAME_AND_TERRITORY - name confirmed by a house-level point inside the same verified territory	0	-
TERRITORY - a house-level point inside exactly one verified territory	0	-
MANUAL - an administrator chose	0	-
Exception (not assigned)	1	-

- **Geocoder precision on these addresses:** AREA 79, PINCODE 51 - none house- or street-level. So **0 deliveries could be assigned by territory**, and the 20 inferred territories were left **unverified** (nothing verifies them automatically: they are hulls of OpenStreetMap anchor points, not surveyed boundaries).
- **Exception reasons:** AMBIGUOUS_MATCH: 1. The delivery that was not assigned is an address listed under two beats; the system suggested a beat and lists both contenders (1 of 1 exceptions has the true beat among them).
- **The previous system** depended on the geocode alone: every point was area- or pincode-level, so 125 of 130 fell into no territory or several, and the 5 it did assign were wrong.
- **A weak geocode never assigns and never overrides a strong name match** - unit-tested and integration-tested on a real PostGIS database.

The fair test

With each delivery's own directory row held out the matcher never sees the answer for the row it judges. Thresholds (score 65 to assign, a lead of 15 points over the next beat) were chosen on this data, but on a wide plateau: **all 85 threshold combinations swept gave zero wrong automatic assignments** (between 88 and 91 correct).



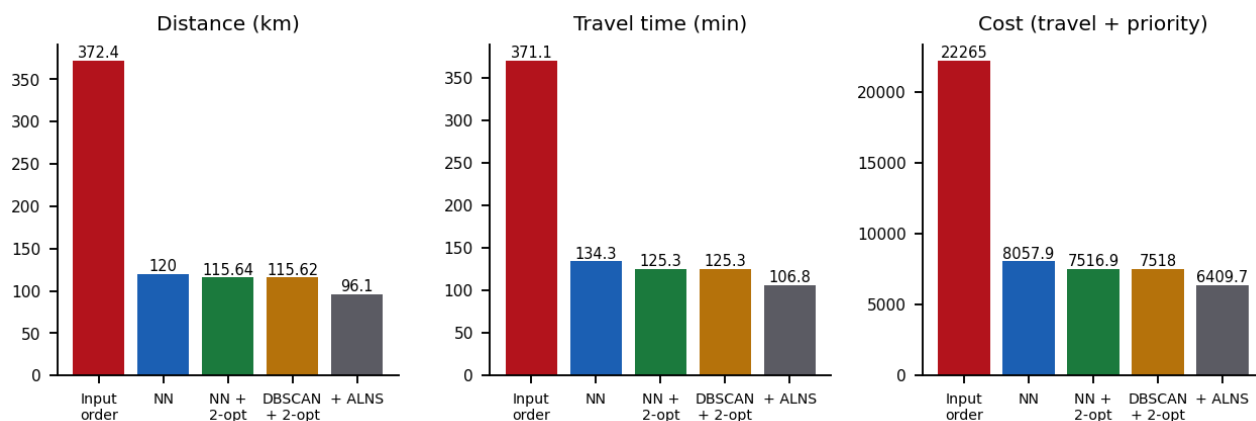
What is not measured: the MEDIUM confidence band was never reached by this data set (covered by unit tests only); the TERRITORY method's fixed confidence (80) is not validated (no house-level ground truth exists); the beat list of the real post office is a PDF that is not parsed, so the directory here is the supplied CSV.

2. Route algorithms

Measured on this machine, with the production ALNS settings (at most 2500 iterations, 1500 ms, 600 without improvement, clusters kept), on the real deliveries with real road times from OSRM, and the same cost function for every stage (driving time + priority; the load term is off because no delivery has a weight). **Read with care:** the geocoder gave these 130 stops only **17 distinct coordinates**, so stops sharing a point are at distance zero; the benchmark ranks the algorithms on ordering the places, not the houses.

all 130 deliveries, one round (130 stops)

Algorithm	Distance	Travel time	Cost	Runtime
Input order	372.4 km	371.1 min	22265	0 ms
Nearest Neighbor	120 km	134.3 min	8057.9	0.8 ms
NN + 2-opt	115.64 km	125.3 min	7516.9	42 ms
DBSCAN + NN + 2-opt	115.62 km	125.3 min	7518	11 ms
DBSCAN + NN + 2-opt + ALNS	96.1 km	106.8 min	6409.7	369.7 ms



ALNS improved on DBSCAN + NN + 2-opt by **14.74 %** in cost (16.88 % in distance) and on plain Nearest Neighbor by 20.45 % (751 iterations, 5 new best routes).

The five smaller rounds (a group of five beats each)

Round	Stops / distinct points	2-opt cost	ALNS cost	ALNS over 2-opt
beats 1-5 (24 deliveries)	24 / 4	2632.2	2632.2	0 %
beats 6-10 (25 deliveries)	25 / 4	2489.1	2489.1	0 %
beats 11-15 (25 deliveries)	25 / 5	3007	3007	0 %
beats 16-20 (25 deliveries)	25 / 4	1706.3	1706.3	0 %
beats 21-25 (25 deliveries)	25 / 4	1799	1799	0 %

ALNS found nothing better than 2-opt on any of these small rounds (4-5 distinct places: 2-opt already reaches the best order, so there is nothing for ALNS to improve). It is reported as it is; ALNS never returns a route dearer than 2-opt, and here it did not need to.

Larger synthetic rounds (model road network - not real roads)

Stops	NN + 2-opt	DBSCAN + NN + 2-opt	+ ALNS	ALNS over 2-opt	Full pipeline runtime
25	77.9 min	77.9 min	72.1 min	0.01 % (cost)	36.7 ms
50	245.7 min	204.0 min	154.5 min	14.66 % (cost)	199.2 ms
100	360.3 min	282.8 min	274.0 min	5.03 % (cost)	784.7 ms
200	542.3 min	440.4 min	385.3 min	8.96 % (cost)	842.0 ms

Deterministic rounds of 25 / 50 / 100 / 200 deliveries with mixed priorities and weights, a street-grid model with a river, and the ALNS wall-clock cap. Travel time is not the cost: with priorities live, the cost also charges lateness of urgent stops, so a route can be

cheaper than another while driving longer.

Keeping DBSCAN clusters together: does it cost route quality? (measured)

Condition	Stops	Drive time, clusters kept	Drive time, clusters free	Difference
Plain (real-data condition)	25 / 50 / 100	90.0 / 123.0 / 184.7 min	89.7 / 123.2 / 184.6 min	-0.3 % / +0.2 % / -0.0 %
Priorities + weights live	25 / 50 / 100	108.0 / 168.1 / 269.0 min	119.2 / 183.9 / 309.9 min	+10.4 % / +9.4 % / +15.2 %

Same drive time with plain deliveries (1.7-4x faster with clusters kept); 9-15 % *shorter* drives with clusters kept when priorities are used (freeing the search lowers the cost by spending driving time). Decision: clusters stay together. Source: `data/bhandup/cluster-preservation-study.txt`.

3. Limitations of these figures

- **Ground truth is the file's layout** (5 deliveries per beat in beat order). There is no independently surveyed answer key.
- **The full-directory accuracy is partly circular**: the supplied beat directory includes each delivery's own main area. Use the hold-out figure (91 of 130 assigned automatically, none wrong) as what the matcher itself achieves; the rest go to an administrator with the true beat among the suggestions.
- **No house-level ground truth** and no house-level geocodes in this data: the geocode-precision logic, the TERRITORY method and address learning are covered by unit and integration tests, not by this accuracy test.
- **One post office, one data set**. Other localities will have other ambiguity; the thresholds were chosen on this set (on a wide plateau) and should be re-validated on new data with `npm run matcher:validate`.
- **Routing on the real deliveries is degenerate** (17 distinct points for 130 stops); the synthetic rounds use a model road network. Neither is a measurement of a real postman's day.
- **Time windows are not modelled** (no data, and the ALNS closed-form insertion cost cannot represent them); **no delivery has a weight**, so the load term of the cost is off for this data.
- The public OSRM demo server and Nominatim were used (fair-use, no uptime guarantee); results depend on what they returned on the day (2026-09-21).